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#include<iostream>

#include<iomanip>

#include<cmath>

using namespace std;

int main()
{
    cout.precision(4);
    cout.setf(ios::fixed);

    int n,i,j;

    cout<<"\nEnter the order of the matrix:\n";

    cin>>n;

    double a[n][n],b[n],c[n],k,eps,y;


    cout<<"\nEnter the elements of matrix row-wise:\n";
    for (i=0;i<n;i++)          //Get the elements of the matrix
        for (j=0;j<n;j++)
            cin>>a[i][j];


    cout<<"\nEnter the initial guess for the eigen-vector:\n";
    for (j=0;j<n;j++)          //Get the iinitial guess for the eigen vector
        cin>>b[j];


    cout<<"\nEnter the accuracy desired:\n";

    cin>>eps;

    k=b[0];                    //Assign some initial value to the eigen value, 'k'

    do

```

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{
    y=k;
    for (i=0;i<n;i++)
    {
        c[i]=0;
        for (j=0;j<n;j++)
            c[i]=c[i]+a[i][j]*b[j];    //After all the iterations axb=c
    }
    k=c[0];
    for (i=1;i<n;i++)
    {
        if(abs(k)<abs(c[i]))
            k=c[i];
    }
    for (i=0;i<n;i++)
        b[i]=c[i]/k;    //Calculate the new Eigen Vector
}while(abs(k-y)>=eps);

cout<<"\n\nThe largest Eigenvalue is: "<<k<<endl;
cout<<"\nAnd the corresponding Eigenvector is: \n";
for (i=0;i<n;i++)
    cout<<b[i]<<endl;
cout<<"\n";
return 0;
}

```